

Dr Andreas Wilzeck

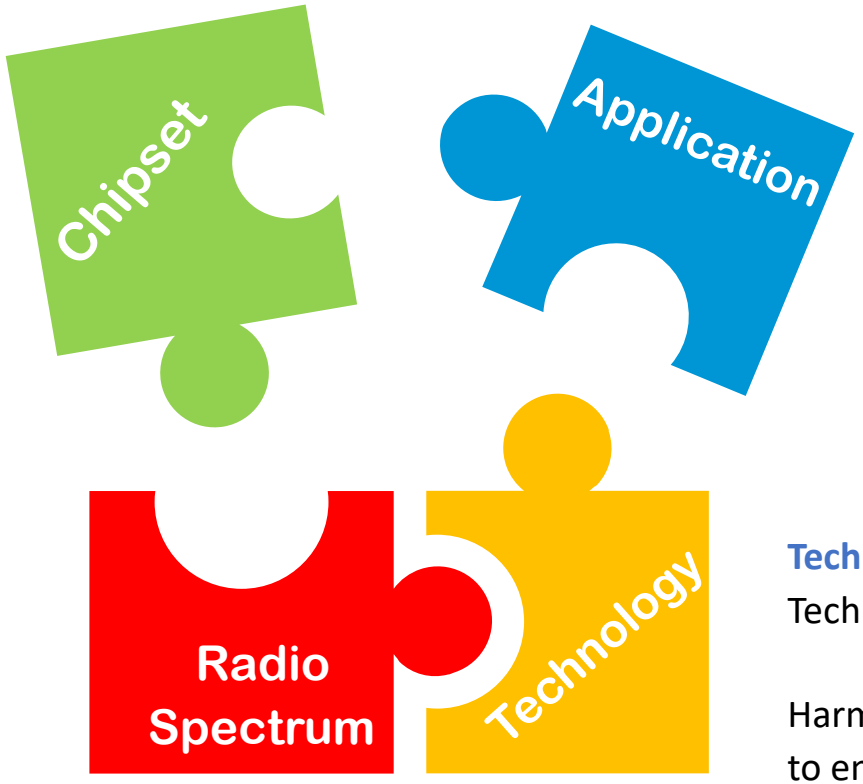
An Introduction to DECT NR+

SENNHEISER

We build the Future of Audio.

What are the requirements for a Wireless Solution?

Economically
viable and scalable
technical implementation of the product.



Application delivering
a value-add to the customer

Access to global radio spectrum
according to
Frequency Regulation
(ITU, CEPT, National)

Technology supports application requirements?
Technical Specification (own, ETSI, 3GPP, IEEE or other)

Harmonised Standard (ETSI)
to enable market access.

DECT NR+

Enabling you and your application.

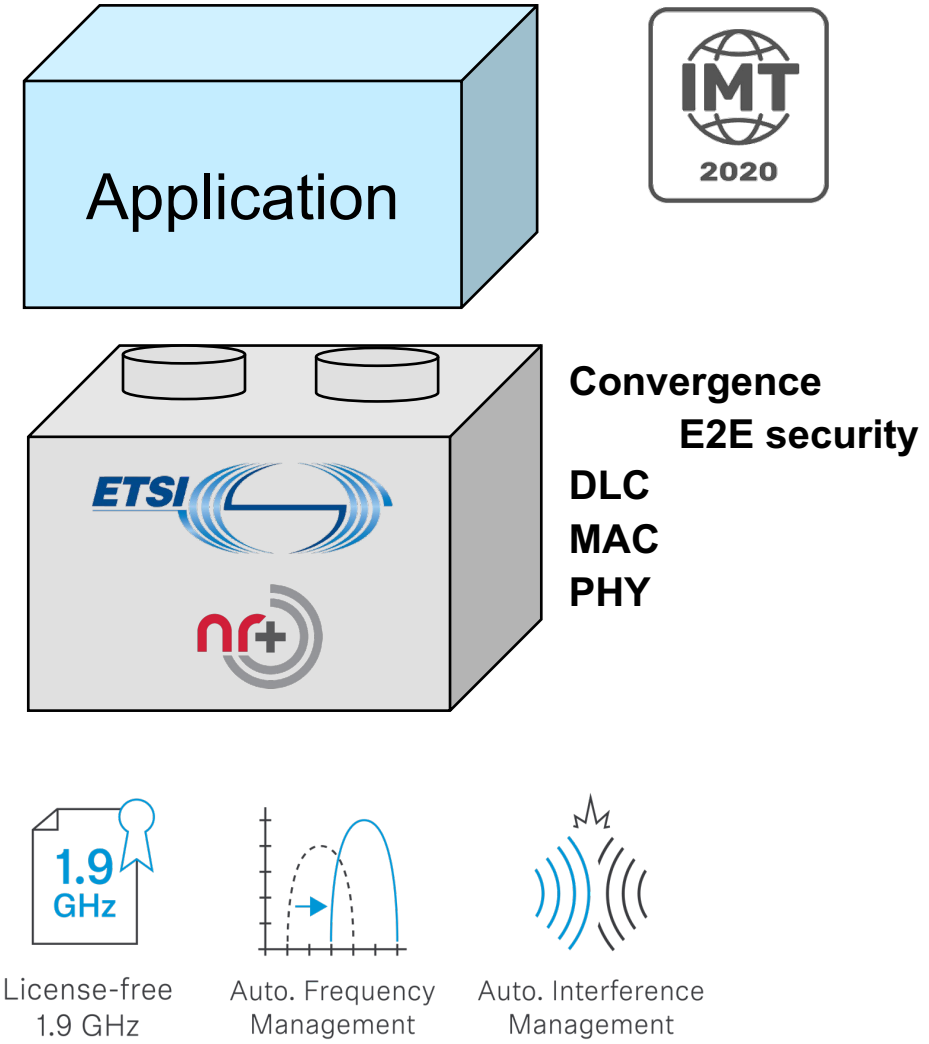
Straightforward innovation path

for many existing use cases

- Evolving the advantages of DECT into the 5G world
- Open Standardization by ETSI as DECT-2020 NR
- Access to additional IMT and non-IMT frequency bands
- No dependence on network operators

Technology enablers

- New flexible PHY/MAC engine based on OFDM TDD TDMA/FDMA including MIMO support
- New network topologies (Point-to-Point, Star, Mesh Tree)
- Designed for operation below 6 GHz, including 1.9 GHz
- Decentralized, autonomous and self-healing



One Technology, Many Local Applications

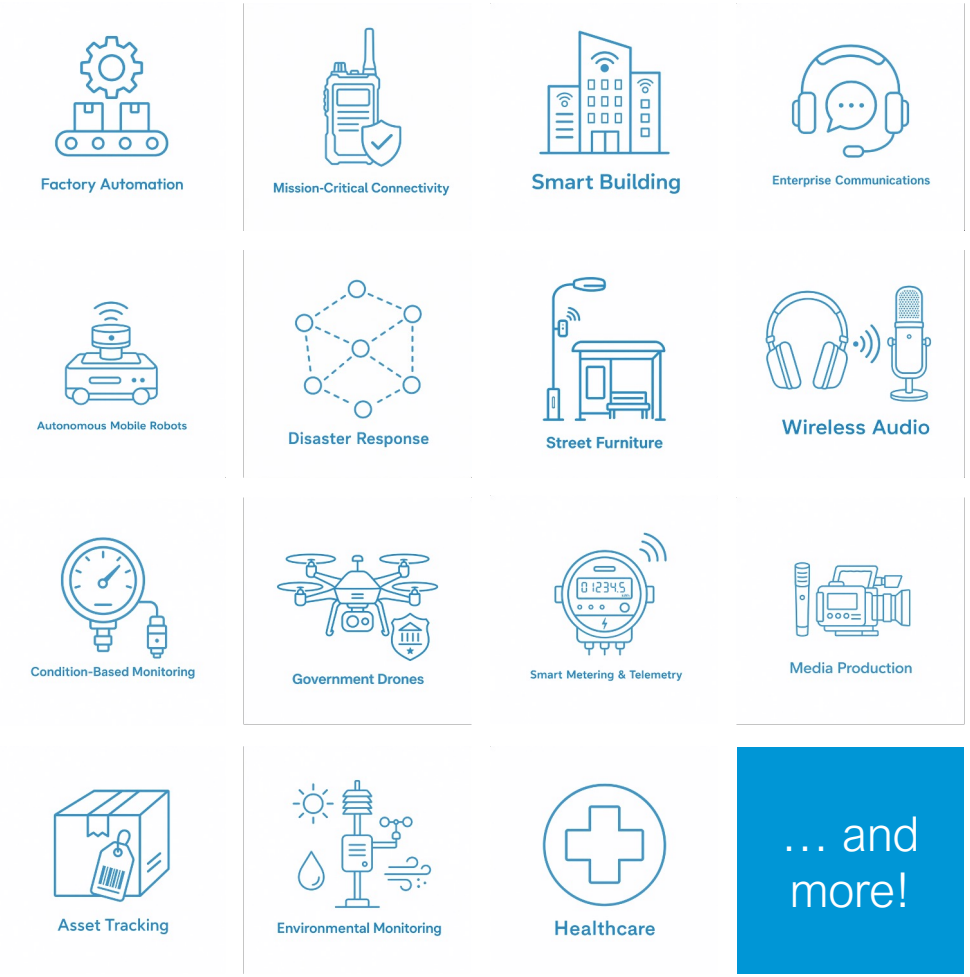
DECT NR+ enables resilient connectivity where it is needed.

Industry & Operations
Factory automation
Autonomous mobile robots
Condition-based monitoring
Asset tracking

Public Infrastructure & Resilience
Disaster-response ad-hoc networks
Mission-critical communications
Public-safety and government drones
Environmental and maritime monitoring

Connected Places
Smart buildings
Smart street furniture
Smart metering and telemetry
Healthcare

Professional Communications
Enterprise communications
Wireless audio
Media production



Core application areas include applications of the System Reference Document on DECT-2020 NR, ETSI TR 103 943

Ecosystem

Technical Standardization Body

- Technical Specifications
- Test Specifications
- Application Profile Specification
- Harmonised Standards
- System Reference Documents

Marketing Organization

- Market Communication
- Regulatory Communication
- NR+ Certification
- Member Communication
- DECT Forum AIR
- Events & Webinars

Chipsets

Stacks

Opener Initiative
Open-Source Reference Stack

Applications

Cordless Telephony
Professional Communications
Wireless Audio

Presentation
Conferencing
Audio for Video
Headset

Intercom
Smart Home/IoT/ Industrial
Smart Grid/Smart Meter

Open Standard, Open Innovation, Open Spectrum?

Open Standard

ETSI TC DECT
Open, consensus-based standardisation of DECT-2020 NR / NR+

IMT-2020 status
A recognised ITU-R IMT-2020 radio-interface technology, designed for mMTC and URLLC applications.

Open Implementation

DECT Forum AIR
connecting research, demonstrators, industry collaboration, funding opportunities and standardisation

Opener Initiative
Apache 2.0 open-source reference implementation of the DECT-2020 NR stack

What this enables
SMEs, universities and local integrators can develop and deploy solutions without relying on a proprietary radio stack.

Open Spectrum

1880-1900 MHz
Established DECT ecosystem band for low-power, local and resilient connectivity

1900-1920 MHz
Potential extension

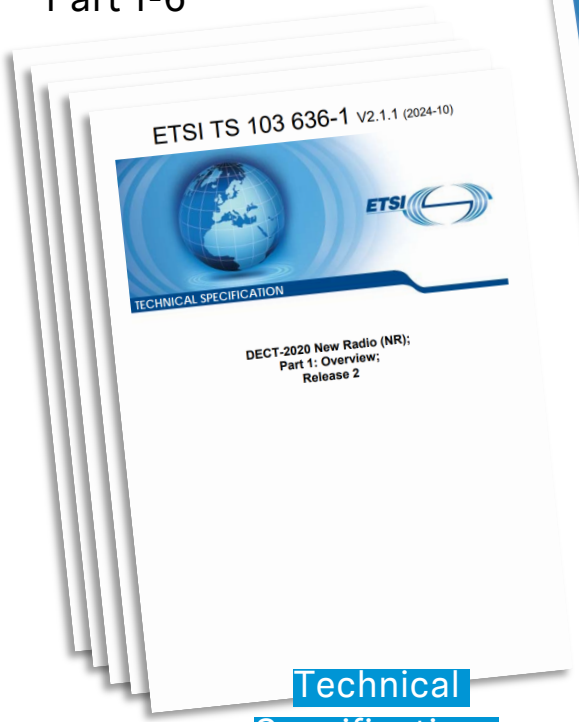
3800-4200 MHz
A potential 400 MHz high-capacity layer for dense local networks, subject to national band plans and incumbent protection

Technical References:
Commission Implementing Decision 2025/2425 and CEPT Report 088

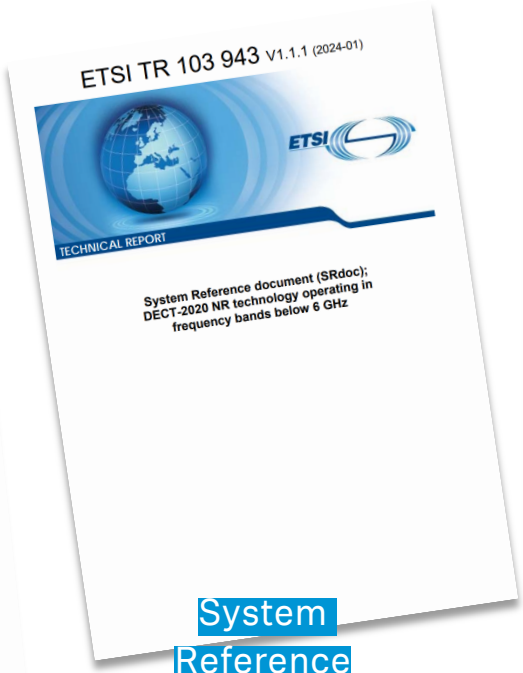
Open standard + open-source implementation + appropriate spectrum = enabling SMEs to innovate locally.

Core Documents

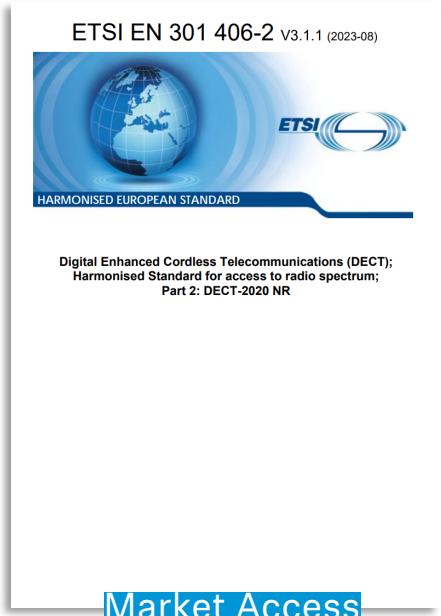
Part 1-6



Technical Specifications



System Reference



Market Access



CEPT Harmonization

An Overview on ETSI DECT-2020 NR Specifications

Technology Specifications
TS 103 636 series

-1 Overview
-2 Radio Transmission and Reception Requirements
-3 Physical Layer
-4 MAC Layer
-5 DLC and Convergence Layer
-6 Security, pre-shared key management
…

Application/Access Profiles
TS 103 874 series

-1 Overview
-2 Smart Metering, City and Buildings
-3 IPv6
-4 Over the Air Software Update Operation
-5 Generic Audio
-6 Industrial Automation
…

Conformance Test Specifications
TS 104 047 series

-1 radio and reception
-2 radio protocols
…

Market Access & Regulatory

EN 301 406-2 Harmonized Standard
TR 103 943 System Reference Document

How DECT NR+ provides design flexibility

How DECT NR+ provides design flexibility

- Shannon-Hartley Theorem:
  Channel Capacity is the theoretical maximum rate at which information can be reliably transmitted over a communication channel.

$$C = B \log_2\left(1 + \frac{S}{N}\right) = B \log_2\left(1 + \frac{S}{N_0 B}\right)$$

Capacity [bit/s] — Signal Power [W] — Bandwidth [Hz] — Noise Power [W] — Noise Spectral Density [W/Hz]

- How can we improve communication performance?
  - Increase B means higher N due $N=N_0B$ → DECT-2020 NR allow to scale bandwidth 1,728 MHz x β with β ϵ {1, 2, 4, 8, 16}
  - Increase S/N means **shorter distance** or **higher transmit power [logarithm!].** → Mesh networking is reducing link distance but adding latency per hop.
  - Increase number of wireless channels via multiple Antennas on Tx and RX means higher complexity. → DECT-2020 NR specifications support multiple antenna schemes.

DECT-2020 NR has a flexible PHY Layer

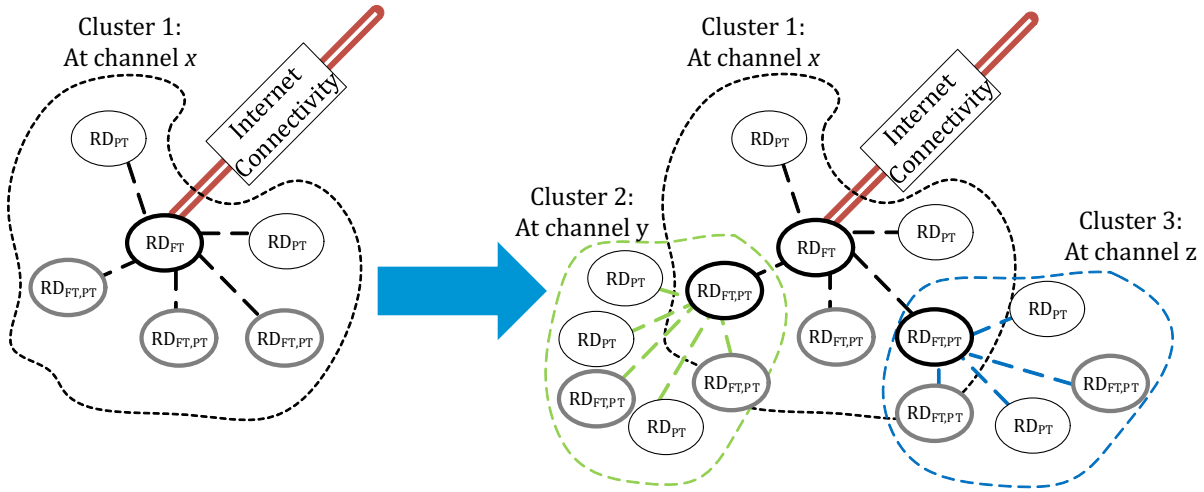
- DECT-2020 NR offers a quite flexible OFDM numerology:
  - Sub-Carrier Spacing (SCS) with μ ϵ {1, 2, 4, 8}:
    - Δf = μ x 27 kHz resulting in 27 kHz, 54 kHz, 108 kHz, 216 kHz
    - **Symbol duration is inversely proportional to subcarrier spacing:** $T_{sym} = 1/\Delta f$
  - Bandwidth Scaling via FFT Size N=β x 64 with β ϵ {1, 2, 4, 8, 16}
- **μ** mainly changes **subcarrier spacing** and thus symbol duration, so it drives **latency and robustness to delay spread**.
- **β** mainly changes **occupied bandwidth** (FFT size / number of subcarriers), so it drives **available payload capacity and data rate**.

Table 4.3-1: Supported transmission numerologies

		β	$B_{DFT}^{\mu,\beta}$ [kHz]	$T_s^{\mu,\beta}$	N_{DFT}^{β}	N_{CP}^{β}	N_{OCC}^{β}	$B_{TX}^{\mu,\beta}$ [kHz]
μ	1	1	1 728	5,7870E-07	64	8	56	1 539
Δ_f^{μ} [kHz]	27	2	3 456	2,8935E-07	128	16	112	3 051
T_{symb}^{μ} [us]	41,667	4	6 912	1,4468E-07	256	32	224	6 075
$N_{symb}^{SLOT,\mu}$	10	8	13 824	7,2338E-08	512	64	448	12 123
$N_{subslot}^{SLOT,\mu}$	2	12	20 736	4,8225E-08	768	96	672	18 171
GI^{μ} [us]	18,52	16	27 648	3,6169E-08	1 024	128	896	24 219
		β	$B_{DFT}^{\mu,\beta}$ [kHz]	$T_s^{\mu,\beta}$	N_{DFT}^{β}	N_{CP}^{β}	N_{OCC}^{β}	$B_{TX}^{\mu,\beta}$ [kHz]
μ	2	1	3 456	2,8935E-07	64	8	56	3 078
Δ_f^{μ} [kHz]	54	2	6 912	1,4468E-07	128	16	112	6 102
T_{symb}^{μ} [us]	20,833	4	13 824	7,2338E-08	256	32	224	12 150
$N_{symb}^{SLOT,\mu}$	20	8	27 648	3,6169E-08	512	64	448	24 246
$N_{subslot}^{SLOT,\mu}$	4	12	41 472	2,4113E-08	768	96	672	36 342
GI^{μ} [us]	20,83	16	55 296	1,8084E-08	1 024	128	896	48 438
		β	$B_{DFT}^{\mu,\beta}$ [kHz]	$T_s^{\mu,\beta}$	N_{DFT}^{β}	N_{CP}^{β}	N_{OCC}^{β}	$B_{TX}^{\mu,\beta}$ [kHz]
μ	4	1	6 912	1,4468E-07	64	8	56	6 156
Δ_f^{μ} [kHz]	108	2	13 824	7,2338E-08	128	16	112	12 204
T_{symb}^{μ} [us]	10,417	4	27 648	3,6169E-08	256	32	224	24 300
$N_{symb}^{SLOT,\mu}$	40	8	55 296	1,8084E-08	512	64	448	48 492
$N_{subslot}^{SLOT,\mu}$	8	12	82 944	1,2056E-08	768	96	672	72 684
GI^{μ} [us]	10,42	16	110 592	9,0422E-09	1 024	128	896	96 876
		β	$B_{DFT}^{\mu,\beta}$ [kHz]	$T_s^{\mu,\beta}$	N_{DFT}^{β}	N_{CP}^{β}	N_{OCC}^{β}	$B_{TX}^{\mu,\beta}$ [kHz]
μ	8	1	13 824	7,2338E-08	64	8	56	12 312
Δ_f^{μ} [kHz]	216	2	27 648	3,6169E-08	128	16	112	24 408
T_{symb}^{μ} [us]	5,208	4	55 296	1,8084E-08	256	32	224	48 600
$N_{symb}^{SLOT,\mu}$	80	8	110 592	9,0422E-09	512	64	448	96 984
$N_{subslot}^{SLOT,\mu}$	16	12	165 888	6,0282E-09	768	96	672	145 368
GI^{μ} [us]	10,42	16	221 184	4,5211E-09	1 024	128	896	193 752

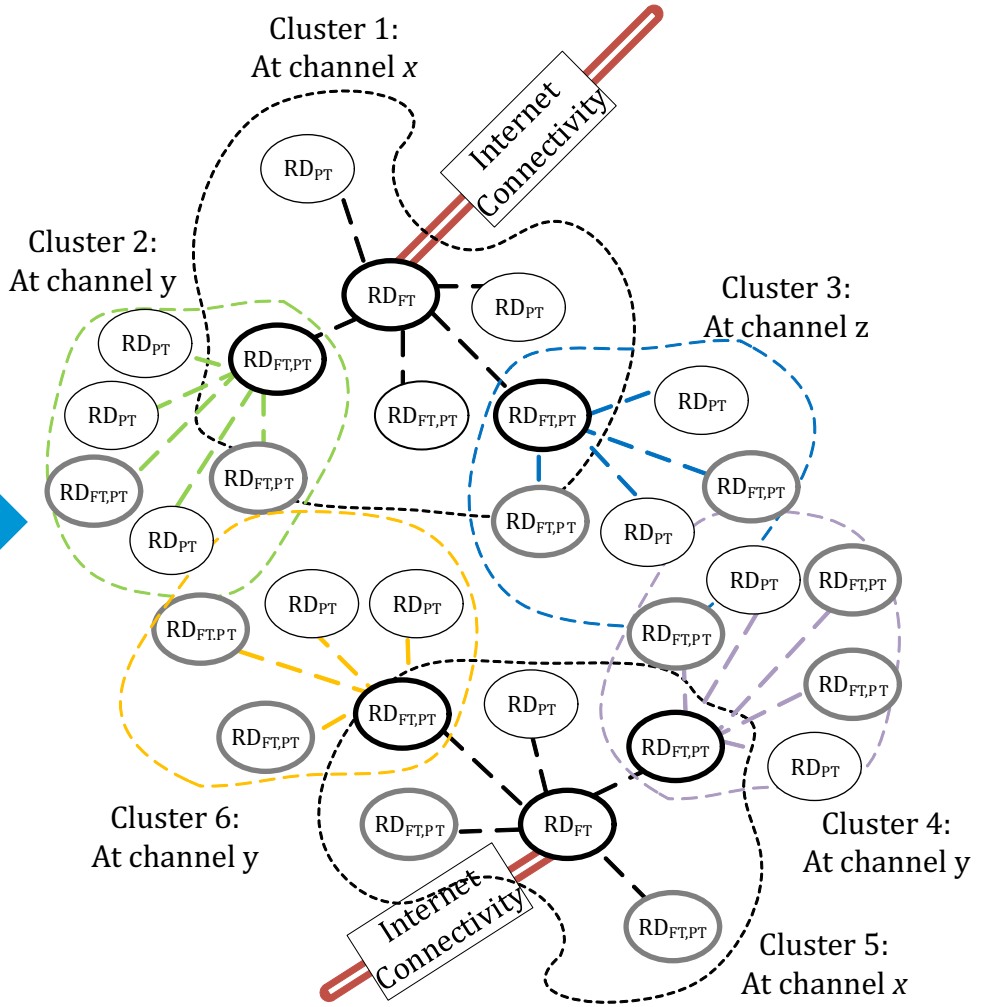
Topology: STAR, Mesh-Tree

RD= Radio Device

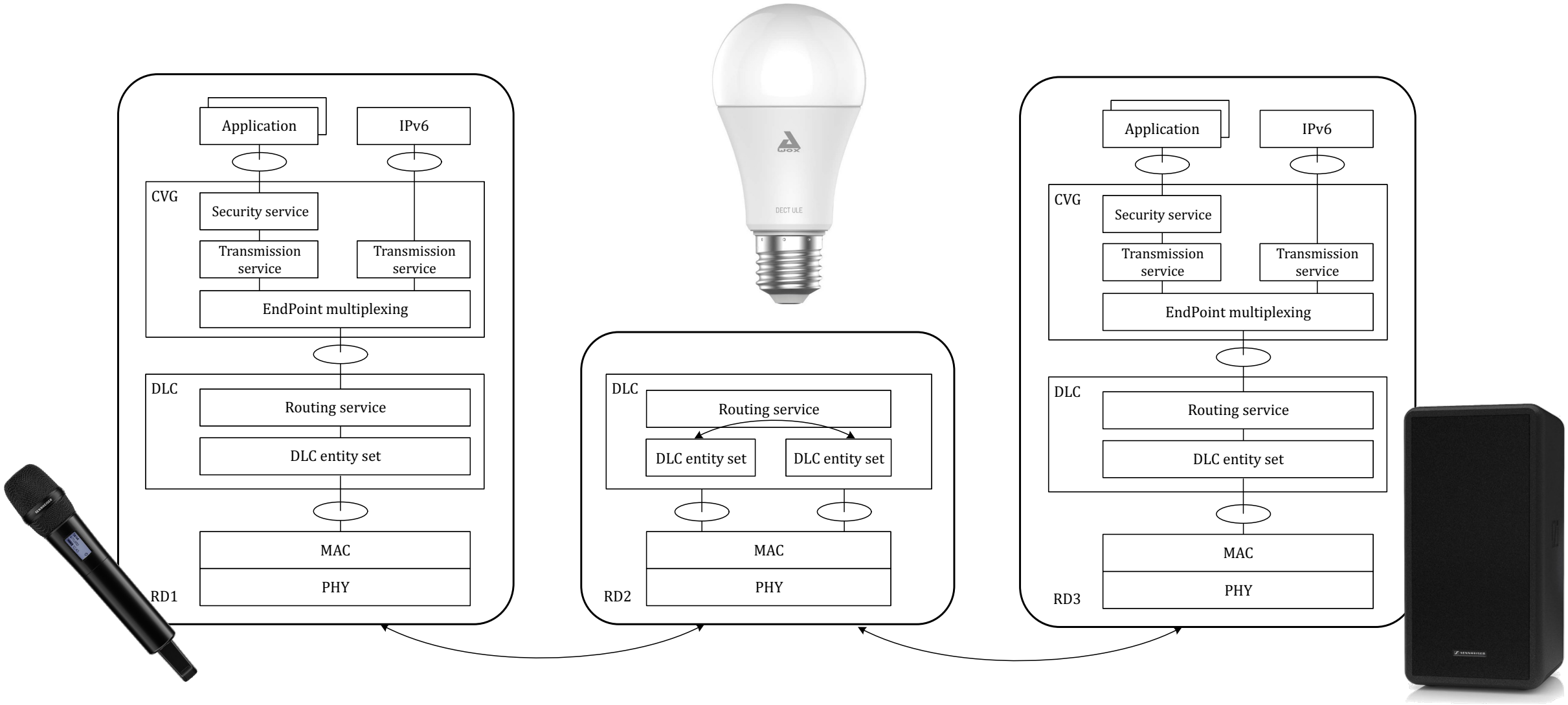


a) Formation of first hop

b) Formation of second hop



Protocol Architecture includes Layer 2 Routing




Thank You.
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We build the Future of Audio.